

Restoring Order to the Deep Sky

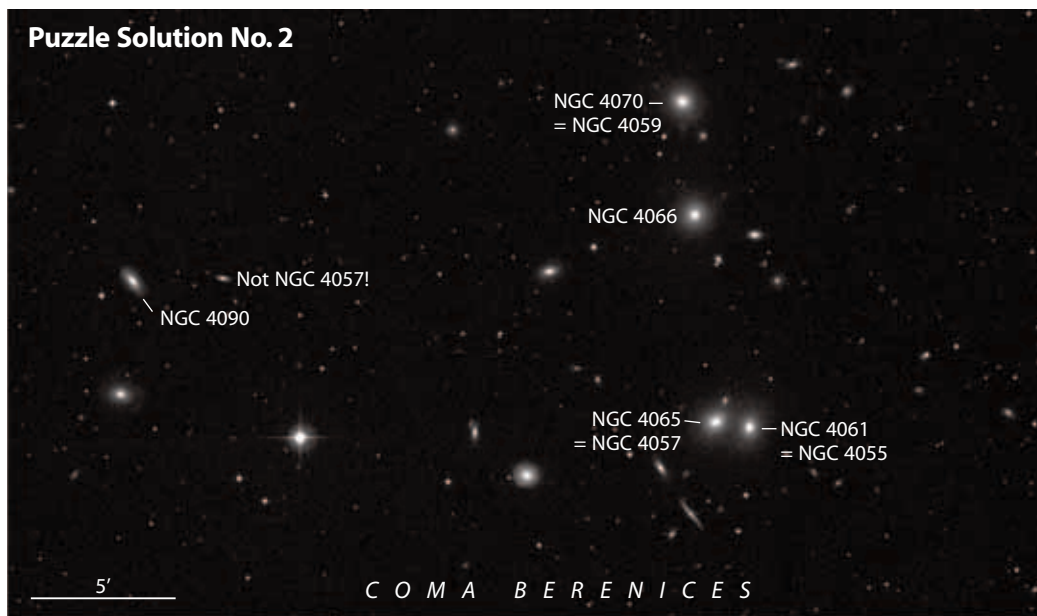
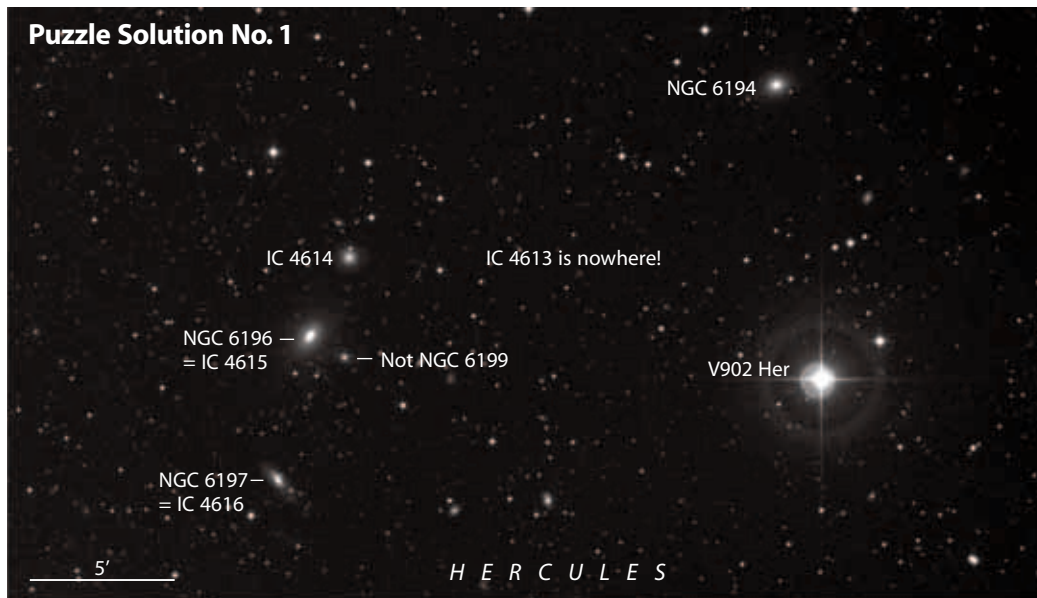
The NGC/IC Project is cleaning up vast numbers of mistaken identities, lost objects, and other mysteries in astronomy's most important deep-sky catalogs. | **By Steve Gottlieb**

LIKE MOST AMATEURS CUTTING their teeth on visual astronomy, I started off with the Messier list and the brightest NGC showpieces, learning my way around the sky and honing my observing skills for a couple of years with a 6-inch f/5 reflector and an 8-inch Schmidt-Cassegrain. After buying a 13-inch f/4.5 Dobsonian, I felt I had the equipment and experience to take on a real challenge — observing all 7,840 entries in the NGC! These objects were all found visually in the late 18th and 19th centuries, so I felt I could retrace the efforts of the pioneer observers using modern equipment and references.

You just looked up the coordinates and data and then star-hopped to the location, right? On a July evening in 1983 at Fremont Peak State Park near Salinas, California, I pointed my 13-inch scope at a faint galaxy group 50' west-southwest of M13 in Hercules. Five galaxies were visible in the area, including three in a distinctive, nearly north-south line (see top photo at right). But when I later checked Sulentic and Tifft's 1973 *Revised New General Catalogue (RNGC)*, I found four listings (NGC 6194, 6196, 6197, and 6199) near the position of the triplet, and I couldn't reconcile their relative positions with my field sketch. Now I was thoroughly confused!

Determined to make some sense of this mess, I wrote to Brian Skiff at Lowell Observatory and Harold Corwin at the University of Texas, Austin, both of whom were experienced in the craft of catalog sleuthing. I learned that John Herschel discovered the brightest galaxy in the area, NGC 6194. The other NGC entries were found by Albert Marth, observing from Malta in 1864 with a 48-inch equatorial reflector — and that's where the problems began.

Marth's positions were offset from the



The author's historical detective work to straighten out a galaxy group in Hercules (top) led him into the NGC/IC Project, which is resolving problems among all 13,226 entries in the NGC and IC catalogs. Galaxies in groups and clusters are especially prone to misidentifications — which are then copied through the decades from one sky atlas or catalog to the next. See page 116 for puzzle number 2. Both images courtesy the Digitized Sky Survey.

true positions by 38 seconds of right ascension and 1.3 arcminutes of declination, far enough that when French astronomer Guillaume Bigourdan began observing the field in 1886 with the 12-inch refractor of Paris Observatory, he mistook faint stars for NGC 6196 and 6197 and then assigned new identities (IC 4615 and IC 4616) and positions to the actual galaxies. Furthermore, he discovered the faintest galaxy in the north-south line (IC 4614), which Marth had missed. But Bigourdan made a new blunder, recording a new faint nebula, IC 4613, where nothing exists.

Based on Bigourdan's accurate metric positions, the three major galaxy catalogs based on the *National Geographic Society-Palomar Observatory Sky Survey* — namely, the *Uppsala General Catalogue of Galaxies (UGC)*, the *Catalogue of Galaxies and Clusters of Galaxies (CGCG)*, and the *Morphological Catalogue of Galaxies (MCG)* — identify the trio as IC 4614, IC 4615, and IC 4616. But when Marth's positions are adjusted for his offset, it's clear that IC 4615 is really NGC 6196 and IC 4616 is really NGC 6197.

What about NGC 6199, the name the

RNGC assigns to an extremely faint galaxy just 1.3' southwest of NGC 6196? Assuming Marth made the same systematic error in his positions, NGC 6199 turns out to be probably just a faint star.

I was hooked by this romp through observational history! From then on, when I returned home from observing, I'd head over to the University of California, Berkeley, astronomy library to dive into the rich 19th-century astronomical literature and check modern identifications against the historical journals.

Just how pervasive are such problems? Once you venture beyond the well-tended data for the brightest NGC objects, you encounter a bewildering maze of contradictory data and misidentifications. A staggering 15 to 20 percent of all NGC entries have known or potential identification problems — poor positions, misidentifications, duplicate entries, incorrect classifications, and confusion with single or multiple stars or even *Palomar Sky Survey* plate defects. Later catalogers have often exacerbated the problem by simply picking the nearest object in the sky when faced with an ambiguous NGC position, instead of inves-

tigating the historical truth. On the amateur end, these errors percolate down into software packages that may misplot or misidentify NGC objects.

The Making of the NGC

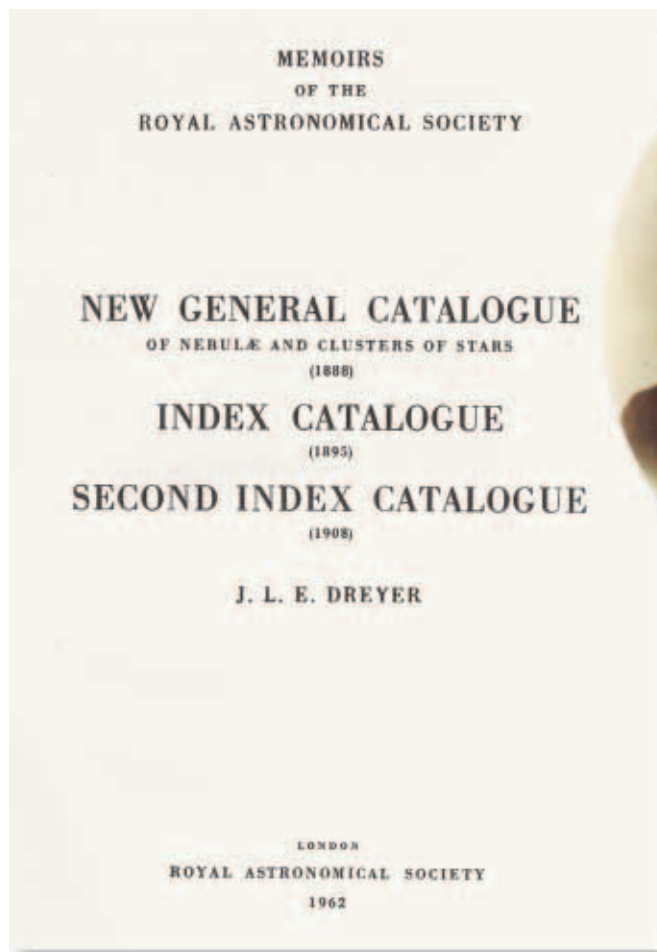
When Johann L. E. Dreyer assembled the *NGC* (the *New General Catalogue of Nebulae and Clusters of Stars*) in 1888, he was faced with correlating precise micrometric positions and rough eyeball positions along with both detailed and vague visual descriptions by different observers using a variety of equipment. Major sources for the *NGC* included William Herschel (using an 18.7-inch speculum-metal reflector in England), John Herschel (18.7-inch speculum-metal reflector, England and South Africa), Lord Rosse and his assistants (72-inch speculum-metal reflector, Birr Castle, Ireland), Albert Marth (48-inch reflector, Malta), Lewis Swift (16-inch refractor, upstate New York), Édouard Stephan (31-inch silvered-glass reflector, Marseilles), Heinrich d'Arrest (11-inch refractor, Copenhagen), and observers at the Leander McCormick Observatory in Virginia (26-inch refractor).

Many of the original Herschel positions had already been checked and improved by later observers, but Dreyer still faced a formidable challenge to compile a coherent catalog. He accomplished this monumental task in slightly more than a year (*S&T*: December 1988, page 621). Working under a publishing deadline, he was unable to research many of the 7,840 entries, and many unconfirmed observations and positions got in. Fortunately, most of the original *NGC* source material can be unearthed at university and observatory libraries today.

What could the original journals and notes reveal after all these years? In many cases they contain essential clues not included by Dreyer, including complete visual descriptions and sketches, micrometer measurements of position offsets from stars, and more. By scouring the rich 18th- and 19th-century astronomical literature, comparing modern data and accurate positions with corrected discovery positions, and making some reasonable extrapolations, many of these cases can be solved. And often a look in the eyepiece today can shed light on what was probably seen more than a century ago.

The NGC/IC Project Is Born

Harold Corwin, a member of the NASA Extragalactic Database (NED) team (<http://>



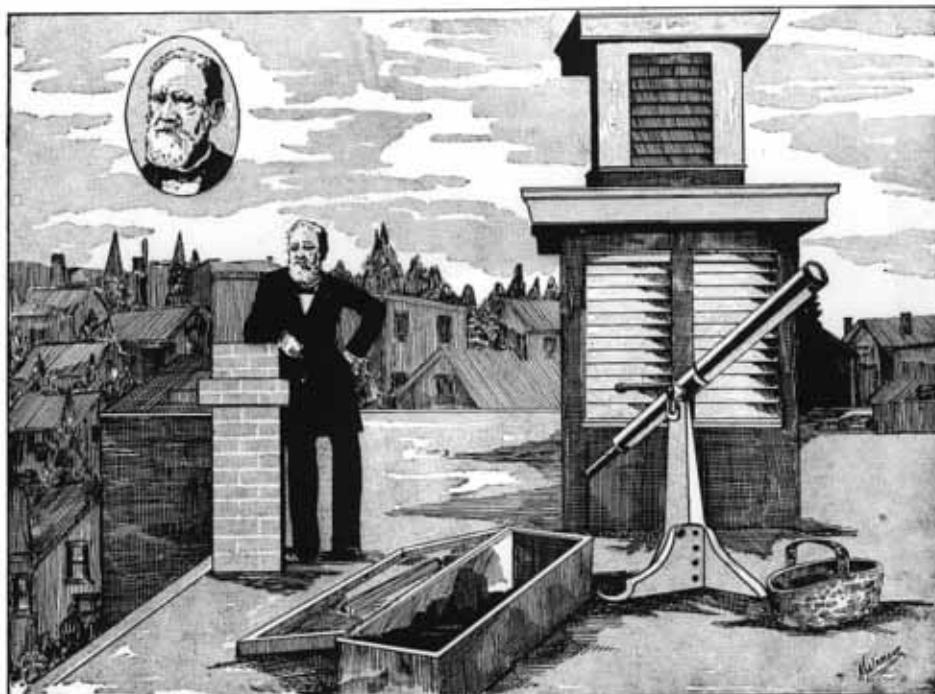
Johann Louis Emil Dreyer (1852–1926) authored the *New General Catalogue of Nebulae and Clusters of Stars*, which is still astronomy's standard list of nonstellar objects 115 years after its publication. With only a year to produce it, Dreyer was unable to research the bonafides of every entry. This title page is from a 1962 reprint.

nedwww.ipac.caltech.edu) and author of several professional galaxy catalogs (*RC3*, *SGC*, *ACO*), has had a special interest in historical observations and catalogs for more than 30 years. Corwin amassed an extensive collection of published and unpublished papers used as source material for the *NGC* and for the two extensions in 1895 and 1908 that constitute the *IC* (*Index Catalogue*).

Out of Corwin's correspondence with other amateur and professional catalog sleuths, a collaborative project evolved with the goal of producing a historically corrected *NGC* and *IC* catalog. Since the mid-1990s, the *NGC/IC* Project has been thoroughly researching all available source material, piecing together the true discovery stories whenever possible, and presenting corrected identifications and modern data for the entire *NGC* and *IC*. Along the way, we've also debugged a number of the errors that plague more modern catalogs.

The following samples from Corwin's *NGC/IC* puzzle-solution files give the flavor of our investigative work and results.

NGC 6, 19, 21, 7831, and 7836. The American amateur Lewis Swift searched for "nebulae" with a 16-inch reflector from



Lewis Swift of Rochester, New York, discovered 560 of the *NGC*'s objects — not with this 4½-inch refractor, but with a 16-inch Clark refractor purchased for him in 1882 by patent-medicine tycoon H. H. Warner following Swift's rise to local fame as a comet discoverer (*S&T*: June 1972, page 364). This illustration, which appeared in the *Rochester Union and Advertiser* for August 13, 1892, was based on a photograph taken in 1880. At right is the market basket in which Swift carried the small telescope's optical parts.

his Warner Observatory in Rochester, New York; he discovered 560 of the NGC entries. On the evening of September 20, 1885, Swift logged several new nebulae about 4° north of Alpha Andromedae. In the case of four of these discoveries, NGC 19, NGC 21, NGC 7831, and NGC 7836, his published positions were systematically off by 1.2^m of right ascension and 8' of declination, with the result that catalogs confused the identifications for the next century!

Applying these corrections to Swift's positions, it's clear that NGC 21 had previously been discovered by William Herschel (as H II 853) and is commonly identified as NGC 29. In addition, Bigourdan was unsuccessful in locating NGC 7831 at Swift's position, but he observed it where it actually lies and assumed it was

a new discovery, so it's now listed in most catalogs as IC 1530.

A fifth object found that evening by Swift, NGC 6, shares the same right-ascension offset as the first four, but this time is off by 45' in declination. The key to unlocking this mystery was found in Swift's original description: "One of 5 stars which point to it is pretty near." An unmistakable chain of five stars trail 4' to the east, and Swift's "pretty near" star is only about 15" east of the galaxy. Because of Swift's poor position, Dreyer was unable to cross-identify other observations, and so this galaxy gained a dual identity as NGC 20.

NGC 4055, 4057, and 4059. John Herschel found these three nebulae in a group on April 29, 1832, and described them simply as "pretty bright." He listed them in

his Slough catalog of 1833 as h1062, h1063, and h1064, but no one has positively identified them in any later catalog!

We find that identification problems abound within galaxy groups and clusters, where accurate positions are essential. Unfortunately, Sir John left us with what he admitted was only a rough polar distance (90° minus the declination) for these three, perhaps accurate to the nearest degree. His description for NGC 4059 also mentions "on meridian with two more" — presumably NGC 4055 and 4057.

Several team members, including Bob Erdmann, Brent Archinal, and Harold Corwin, put their heads together to tackle this problem. They compared positions and descriptions with the other four cluster members that Herschel found during the same sky sweep. He gave about the same polar distances for these as for the three questionable objects, indicating that the declinations of all should be similar; Herschel's sweeps were fairly limited in declination. And based on his description "pB" (pretty bright), the three problematic objects should be among the brighter galaxies in the group.

The four brightest members of the group are NGC 4061, 4065, 4066, and 4070 (see the lower photo on page 113). But Herschel clearly observed NGC 4066 during his sweep, leaving NGC 4061, 4065, and 4070 (previously discovered by John's father, William) as the possible candidates. As another clue, Herschel gave more precise positions for NGC 4055, 4057, and 4059 in his *General Catalogue* published in 1864, and his relative positions are a reasonable fit to these galaxies. Yet several modern catalogs and atlases reassign the name NGC 4057 to PGC 38278, an extremely faint galaxy 3' west of NGC 4090.

Pulling these clues together, our investigation suggests that NGC 4055 is in fact NGC 4061, NGC 4057 is NGC 4065, and NGC 4059 is NGC 4070.

NGC 4153. A famous mystery, long unsolved, involves this object. William Herschel noted the nebula on February 15, 1784, only two months after he started observing with his first "20-foot" (focal length) telescope. He placed it 1.5^m of right ascension preceding (west) and 2.2° south of 5 Comae Berenices and described it as "B, pL, IE, bM, m" (bright, pretty large, little extended, brighter middle, milky). Although this sounds like a fairly prominent object, there is nothing at his offset position from the star, and NGC 4153 has not been identified since.

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Dr. J. L. E. DREYER, *A New General Catalogue*

No.	G. C.	J. H.	W. H.	Other Observers	Right Ascension, 1860.0	Annual Proper Motion, 1860.0	North Polar Distance, 1860.0	Annual Proper Motion, 1860.0	Summary Description	Notes
6181	4218	...	II 753	Schultz	16 26 14	+2.62	69 52.4	+7.9	pB, pL, vIE, pgmbM	
6182	4219	...	III 813	...	16 26 50	1.26	34 9.9	7.9	vF, vS, IR	
6183	4220	3639	...	...	16 26 50	6.31	159 4.9	7.8	vF, vS, R, gbM	
6184	3822	...	...	St VII	16 26 53	2.01	49 8.0	7.9	vF, vS, R, vlbM	
6185	4221	1965	...	...	16 28 15	2.19	54 21.7	7.8	F, S, R, gbM, * 11 sp	
6186	4222	...	...	III 730	16 28 19	2.58	68 9.9	7.7	vF, vS, IE	
6187	...	...	...	Young	16 28 58	1.10	31 59.3	7.7	vF, vS, lrbM	
6188	4223	3640	...	...	16 29 15	4.48	138 44	7.6	IE, vL, vIE, B w inv	
6189	...	...	...	Sw II	16 29 21	0.95	30 5.3	7.7	vF, pS, IE	
6190	...	...	...	Sw II	16 29 34	1.04	31 15.9	7.7	vF, pS, R, F w ne	
6191	...	...	...	Sw IV	16 29 35	1.02	30 54.3	7.7	pF, pL, E, 2 st p	
6192	4224	3641	...	A 483	16 30 29	4.23	133 5.3	7.5	Cl, pL, pRI, IR, at 11...14	†
6193	4225	3642	...	A 413	16 30 52	4.47	138 29.1	7.5	Cl, vL, IRI, IC, rrr, F nob inv	
6194	4227	1967	...	Schultz	16 31 35	2.16	53 30.9	7.5	vF, vS, sbM * 12	
6195	4226	1966	III 893	...	16 31 44	2.06	50 41.5	7.5	vF, S, R, gbM, bet 2 st	
6196	5823	...	...	m 311	16 32 13	2.16	53 37	7.4	vF, vS, stellar	
6197	5824	...	...	m 312	16 32 18	2.16	53 42	7.4	vF, E, stellar	
6198	...	...	...	Sw IV	16 33 8	1.10	32 12.8	7.4	vF, vS, R, 2 st f	
6199	5825	...	...	m 313	16 33 50	2.16	53 38	7.3	vF	
6200	4228	3643	...	...	16 33 50	4.42	137 11.8	7.2	Cl (in Milky Way)	
6201	5826	...	...	m 314	16 34 22	2.52	65 58	7.2	vF, vS	
6202	...	...	...	Sw IV	16 34 32	0.71	27 45.5	7.3	aeF, pS, * f	
6203	5827	...	...	m 315	16 34 34	2.52	65 57	7.2	vF, vS	
6204	4229	3644	...	A 442	16 36 7	4.40	136 45.6	7.1	Cl, pRI, eiCM, at 11...12	†
6205	4230	1968	...	{Halley 1714, M 13}	16 36 40	2.14	53 16.3	7.1	!! F, vB, vRI, vgcCM, at 11...	†
6206	...	...	...	Sw V	16 37 25	1.00	31 7.1	7.0	pF, vS, R, stell, 3 vF st ne	
6207	4231	1969	II 701	...	16 38 6	2.13	52 54.1	7.0	pB, pL, E 45" ±, vgmM	
6208	4232	3646	...	A 364	16 38 18	4.78	143 33.5	6.9	Cl, L, RI, iCM, at 9...12	
6209	4233	3645	...	...	16 38 24	7.03	162 20.5	6.8	vF, pL, vglbM	
6210	4234	1970	...	M 5, LL 30510	16 38 38	2.51	65 56.2	6.9	Cl, vL, vS, R, disc & border	†
6211	...	...	...	Sw VI	16 38 38	1.06	31 55.5	6.9	vF, pS, R, sp of 2	
6212	5828	...	...	St II	16 38 41	2.02	49 55.8	6.9	vF	
6213	...	...	...	Sw VI	16 38 53	1.06	31 54.3	6.9	aeF, vS, R, nf of 2	
6214	...	...	...	Sw I & IV	16 38 54	0.22	23 41.5	6.9	aeF, vS, R	
6215	4235	3647	...	...	16 39 4	5.15	148 44.8	6.8	pF, R, vglbM, * 4 p 79"	
6216	4237	3648	...	A 454	16 39 20	+4.31	134 28.3	6.8	Cl, pS, pRI, pC, at 12...15	
6217	4236	...	I 280	G Rümker	16 39 25	-3.04	11 31.3	7.0	R, cl, IE, sbM	

A page of the NGC. The first column gives the NGC serial number, assigned to all objects in order of right ascension for 1860.0. The next three columns list their numbers in John Herschel's *General Catalogue* or Dreyer's supplement to it, additional lists of John Herschel's, and William Herschel's catalog.

The closest reasonable object for the missing nebula is another early discovery of Herschel's, NGC 4147 (his H I 19), a small globular cluster in Coma Berenices. Herschel did note it just a month after NGC 4153, and his description is quite similar: "vB, pL, gbM" (very bright, pretty large, gradually brighter toward the middle). At the time when William Herschel noted NGC 4153, he was still improving his method of determining positions. Although this identification is certainly no slam-dunk, it's reasonable to assume that the February observation of NGC 4153, earlier on the "learning curve," actually refers to NGC 4147.

IC 2120 and a comet. The *Index Catalogues* have their own mysteries. One of our most enjoyable sleuthing chases involved IC 2120, found by Bigourdan at Paris Observatory. Since the late 1940s IC 2120 has been identified with Minkowski 2-3, located at $5^{\text{h}} 18^{\text{m}} 10.3^{\text{s}}$, $+37^{\circ} 33' 28''$ (2000.0 coordinates). This object was originally cataloged as a planetary nebula (PK 169-0.1) based on its morphology, but radio and infrared observations indicate that it is likely a compact H II region (a diffuse nebula of ionized hydrogen). But

there's a problem. This faint, strongly reddened object is located $39'$ south-southwest of Bigourdan's discovery position, and it is so dim that it was unlikely to have been picked up visually in any case. So what did Bigourdan see?

Digging out Bigourdan's notes, we find that he first assumed the object was Comet 113P/Spitaler (1890), which he had observed just an hour and a half earlier; he returned to the field apparently intending to reobserve the comet. But he thought he was seeing something else. He noted that the object, which was eventually designated IC 2120, was

John Herschel (1792–1871) published his monumental *General Catalogue of Nebulae* in 1864; it included positions of 5,079 nebulae and clusters, mostly discovered by him or his father William. When Dreyer overhauled and extended it as the *New General Catalogue*, he modestly urged astronomers to keep using the Herschel designations for objects rather than his own NGC numbers. "It is with much regret that I found it necessary to introduce new numbers," Dreyer wrote in the *NGC's* introduction, "and it is greatly to be hoped that these will be quoted as little as possible."



COLLECTION OF CRAIG MICHAEL LUTTER

“clearly nebulous and of magnitude 13.4–13.5 and distinguished with difficulty from a very close neighboring star.” Fortunately, Bigourdan left us with accurate micrometric offsets from his comparison star. The comet might provide the clue. Where exactly was it located?

Brian Marsden (Harvard-Smithsonian Center for Astrophysics) provided us with an accurate ephemeris for the comet, and it turned out that Bigourdan’s precise offsets for the “nebula” fall within one arc-second of the position of the comet at the time of his observation. So IC 2120 is Comet 113/P Spitaler, which Bigourdan did not realize he had reacquired. This is the only confirmed case where an NGC/IC object refers to a comet.

The NGC/IC Project Today

Currently available on the NGC/IC Project site, www.ngcic.com, are summaries of several thousand of these puzzle solutions by principal investigators Harold Corwin and Malcolm Thomson (click on “Corrected NGC/IC”). German amateur Wolfgang Steinicke has compiled a *Revised NGC/IC* with precise positions and extensive catalog data, along with a table of biographical information on the original 160 contributors to the *NGC* and *IC*. An interactive NGC/IC observing-list generator, provided by webmaster Bob Erdmann, is available for planning observing sessions.

And as for that ambitious observing project I began 20 years ago? I’ve contributed visual descriptions of more than 6,000 NGC objects, and in many cases these have helped to pin down the correct identifications.

Has all this effort been worthwhile? The NGC and IC designations are still the most commonly used by both professionals and amateurs. Cleaning up the *NGC* and *IC* provides many benefits from both a pragmatic point of view and a historical perspective; proper attribution of discoveries is an essential thread in maintaining the integrity of astronomical history. For me, the allure of the *NGC* and *IC* is that they connect us through the chain of astronomical history to the early observers who first struggled to understand the mysterious deep-sky universe. You’re invited to visit the NGC/IC Project site and experience that historical connection yourself.

STEVE GOTTLIEB is a regular contributor to *Sky & Telescope*. His most recent article, “Celestial Odd Couples,” appeared in the July issue. He can be reached at sgottlieb@telis.org.